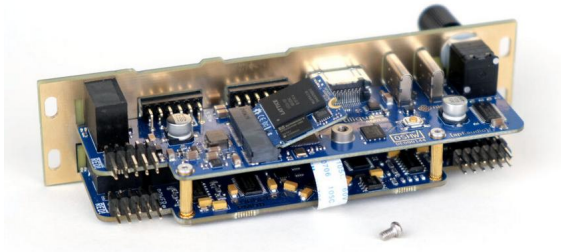


tiliqua: FPGA-based audio DSP platform

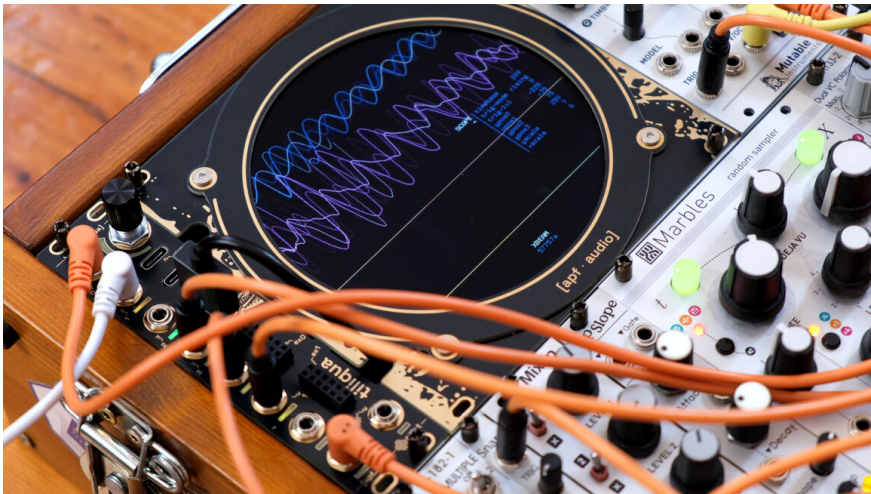
ORConf 2025

github.com/apfaudio/tiliqua

Seb Holzapfel / seb@apf.audio



Eurorack?



- ▶ Standard format for building synthesizers.
- ▶ **Eurorack ecosystem:** > 15,000 modules, > 1,000 different manufacturers.

What is tiliqua?



- ▶ Open hardware platform for audio & video synthesis (Eurorack module)
- ▶ Open gateway library of DSP components (Amaranth HDL)
- ▶ Easy to share designs – debugger built in, hot bitstream switching.
- ▶ Useful to musicians, whether they touch HDL or not.

4x IN
(192kHz, DC-coupled)

4x OUT
(192kHz, DC-coupled)



encoder

JTAG adapter

USB2 high-speed

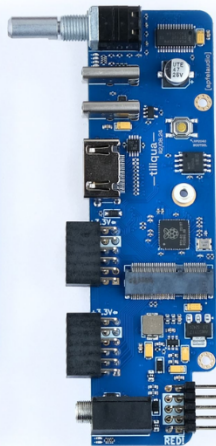
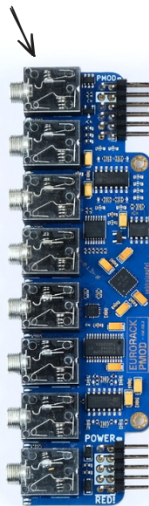
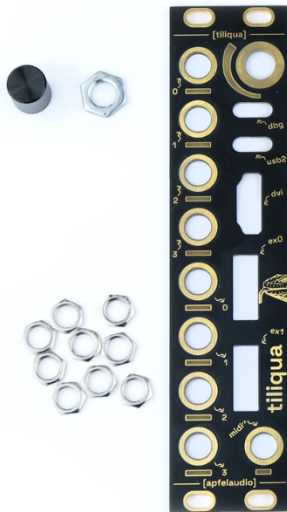
display output

2x expansion PMODs

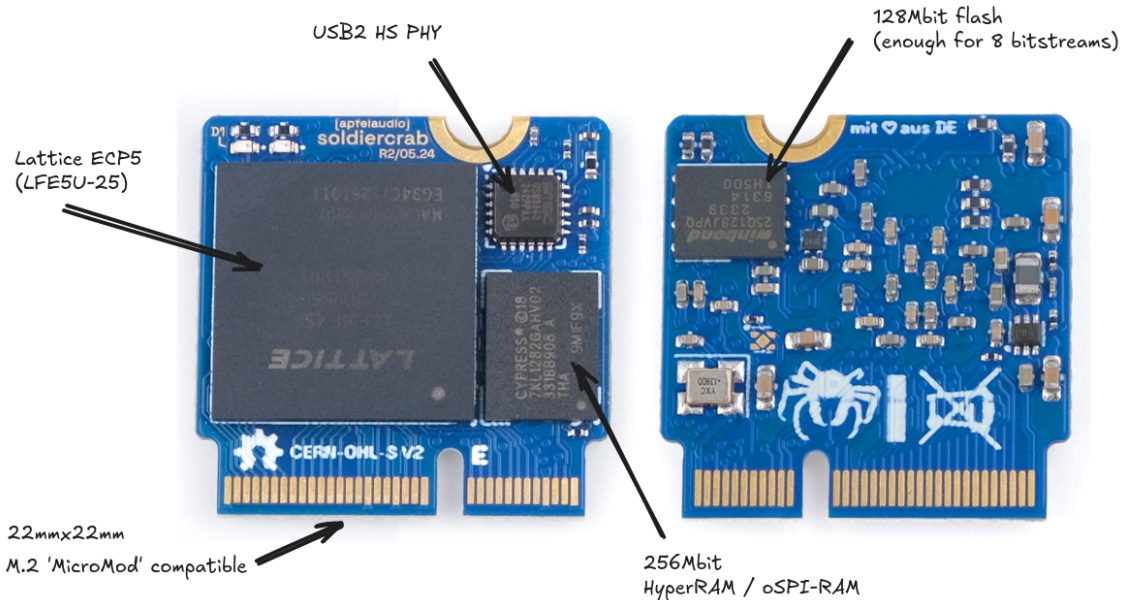
MIDI TRS IN

Audio Interface

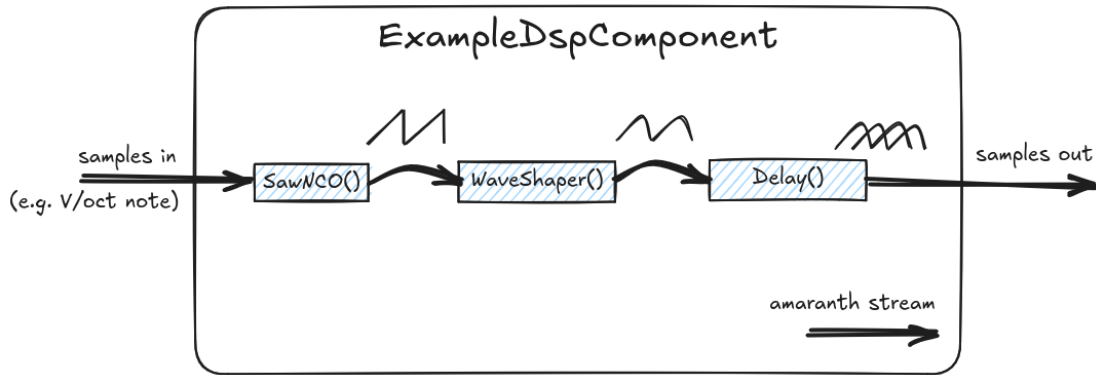
"SoldierCrab" ECP5 SoM



Tiliqua Motherboard



DSP pipelines, Amaranth HDL and streams



New things!

- ▶ **More DSP cores, lib.fixed** – FFTs, STFTs, frequency-domain stuff!
- ▶ **Dynamic pixel clocks** – hot-swap screens.
- ▶ **USB2 Host** – in pure gateware, based on LUNA.
- ▶ **Persistent options** – bitstream auto-restore on power loss.

tiliqua DSP library - frequency-domain processing (new!)

- ▶ **FFT** – forward + inverse, iterative, low area usage.
- ▶ **CORDIC** – complex/polar conversions.
- ▶ **Window** – windowing core and selectable window functions.
- ▶ **STFT** – forward + inverse, frequency-domain processing of time-domain signals.
- ▶ **SpectralEnvelope** – compute filtered power spectra over time.
- ▶ **SpectralCrossSynthesis** – vocoder-like effect.

Also: check out Amaranth lib.fixed

- ▶ **Extension to Amaranth standard library**
- ▶ **RFC + Implementation (WIP):** github.com/amaranth-lang/amaranth/pull/1578
- ▶ **Comments welcome!**

Dynamic clocking (new!)

- ▶ People have different screens (pixel clock).
- ▶ Same bitstream should work everywhere.
- ▶ Core reads display EDID, dynamically adapts resolution!
- ▶ Details: <https://github.com/apfaudio/tiliqua/pull/107>

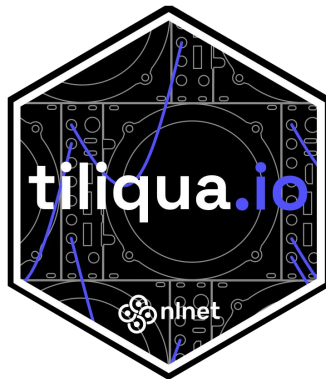
USB2 Host (new!)

- ▶ Cheapest MIDI keyboards only have USB MIDI :(
- ▶ Solution: pure-gateway USB2 high-speed host. (first?)
- ▶ Details: <https://github.com/apfaudio/tiliqua/pull/65>
- ▶ Builds on LUNA: <https://github.com/greatscottgadgets/luna>

Persistent Options (new!)

- ▶ Wear-levelled persistent storage to configuration flash
- ▶ Store last bitstream, all options, autoboot.
- ▶ Details: <https://github.com/apfaudio/tiliqua/pull/129>

Thank you... Funding!



Thank you... Preorders!

CROWD SUPPLY

BROWSE

APPLY

ABOUT

Search

Tiliqua

A powerful, hackable FPGA-based audio multitool for Eurorack



\$46,763 raised
of \$12,000 goal

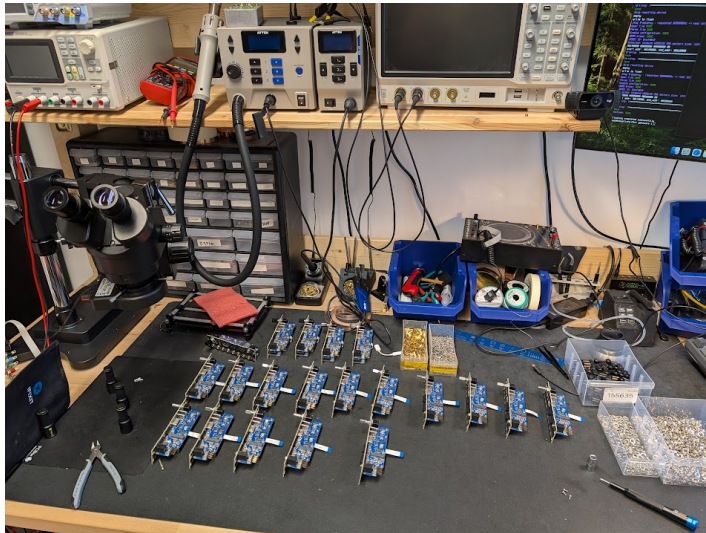
389% Funded!

6
updates

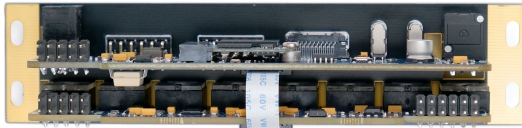
Jun 19
funded on

Available for pre-order.

Production!



Thank you for listening!



- ▶ contact: seb@apf.audio
- ▶ repository: github.com/apfaudio/tiliqua
- ▶ mailing list: apf.audio

want one?

- ▶ crowdsupply.com/apfaudio/tiliqua

(bonus) Hardware / EMC: chamber



- Open source TEM Cell design (By Petteri Aimonen: essentialscrap.com/tem_cell)

(bonus) Hardware / EMC: fail



(bonus) Hardware / EMC: what worked

- ▶ Reroute Switchmode supplies
- ▶ Reduce FPGA pin drive strengths
- ▶ Reduce/remove split ground planes
- ▶ **Careful with long cables!**
- ▶ Spread-spectrum PLL (on ECP5!)

